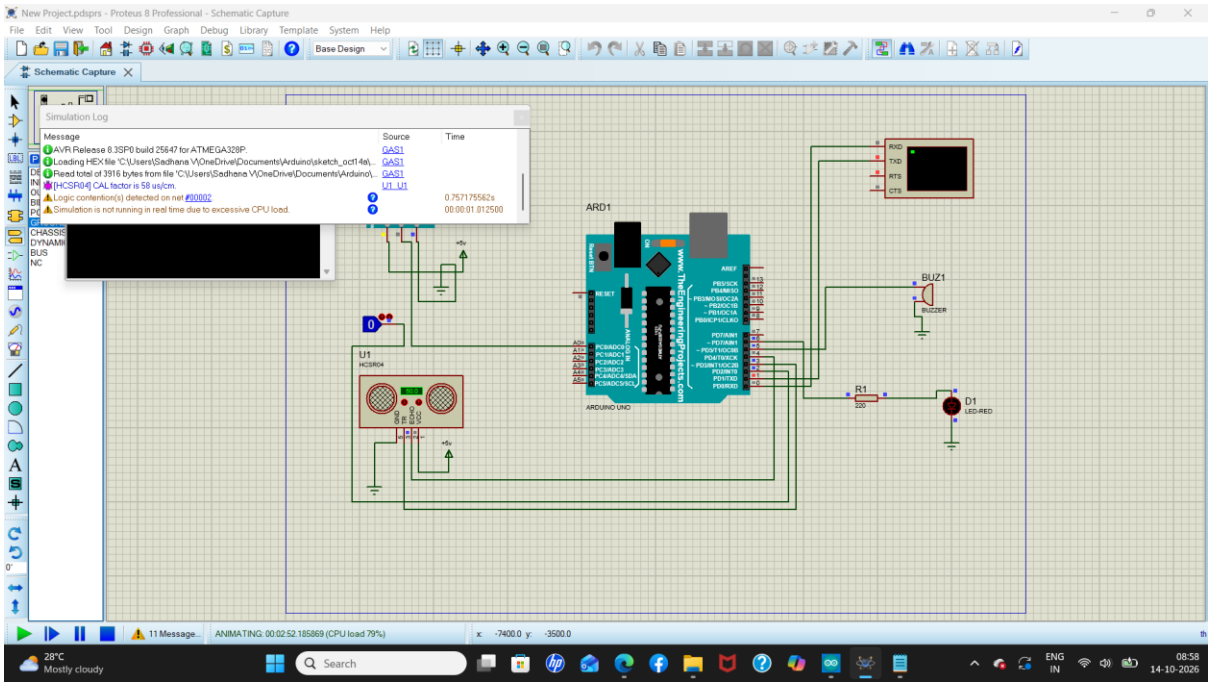
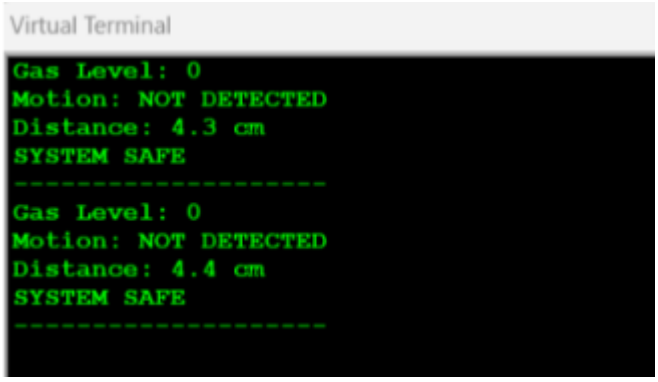
EXP-1 Arduino UNO-Based Smart Home Safety and Occupancy Monitoring System using Gas, PIR and Ultrasonic Sensors

Screenshot:





Source code:

```
#define MQ2_PIN A0

#define PIR_PIN 2

#define TRIG_PIN 9

#define ECHO_PIN 10

#define LED_PIN 13


int gasValue = 0;

int gasThreshold = 400;

int pirState = 0;
```

```
long duration;
float distance;

void setup() {
  pinMode(MQ2_PIN, INPUT);
  pinMode(PIR_PIN, INPUT);
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);

  Serial.begin(9600);

  Serial.println("SMART HOME SAFETY SYSTEM");
  Serial.println("------------------------");
  delay(2000);
}

void loop() {

  // Read MQ-2 Gas Sensor
  gasValue = analogRead(MQ2_PIN);

  // Read PIR Motion Sensor
  pirState = digitalRead(PIR_PIN);

  // Ultrasonic Sensor Measurement
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG_PIN, HIGH);
```

```
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);

duration = pulseIn(ECHO_PIN, HIGH, 30000);

if (duration == 0) {
  distance = -1;
} else {
  distance = duration * 0.0343 / 2;
}

// Gas Leakage Detection
Serial.print("Gas Value: ");
Serial.println(gasValue);

if (gasValue > gasThreshold) {
  Serial.println("WARNING: GAS LEAKAGE DETECTED!");
  digitalWrite(LED_PIN, HIGH);
} else {
  Serial.println("Gas Level: NORMAL");
  digitalWrite(LED_PIN, LOW);
}

// PIR Motion Detection
if (pirState == HIGH) {
  Serial.println("Occupancy: PERSON DETECTED");
} else {
  Serial.println("Occupancy: NO MOTION");
}

// Ultrasonic Distance Measurement
```

```
  if (distance >= 0) {
    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.println(" cm");

    if (distance < 50) {
      Serial.println("Object Detected Nearby");
    }
  } else {
    Serial.println("Distance: No Echo");
  }

  Serial.println("------------------------");

  delay(1000);
}
```